Generative AI Academy

Program flow

- Day 1:
  - EDA : Data Quality checks, Data insights, well articulated, deep insights
    - Structured – JSON, nested JSON, SQL
    - Unstructured: text
    - Data preprocessing with python: lists, dictionaries
  - Graphs and Plots
    - Rich and meaningful visual analysis and insights with matplotlib and seaborn
  - Azure Machine Learning
    - EDA on Azure ML

- Day 2:
  - Knowledge Graph deep dive
  - Prompt Engineering refresh

- Day 3:
  - Integration: Graph DB, Vectors DB, SQL, hybrid mechanisms
  - Mini capstone – to be presented by 12/13 Jan

Data Quality & Data Statistics for Structured and Unstructured Healthcare Data

Why Data Quality Matters in Healthcare

Clinical safety & decision accuracy

Regulatory & audit requirements

ML / GenAI sensitivity to bad data

“Garbage in → hallucinations out”

Data quality is risk management, not cleanup.

Healthcare data landscape

Structured:

claims, billing, demographics

Semi-structured:

JSON, nested JSON, FHIR

Unstructured:

clinical notes, reports

Reality: Most modern healthcare data is nested JSON + text

Typical healthcare data

Example payload:

Patient → Demographics
Encounter → timestamps
Vitals → nested objects
Conditions → arrays

Challenges

Optional fields
Deep nesting
Mixed types
Schema drift

Common data quality issues

- Missing critical fields (timestamps, vitals)
- Wrong data types ("120" vs 120)
- Out-of-range clinical values
- Duplicate patients / encounters
- Inconsistent nested structures

Matplotlib and seaborn

Matplotlib
Draws lines, shapes, text, axes, ticks, legends

- Graphics Engine
- Foundational
- Highly customizable but needs low level code
- Steeper learning curve

Seaborn
Takes a dataset and plots nicely using matplotlib

- Built on top of matplotlib
- Beautiful default styles
- Built in support for stats plots
- Integrates easily with pandas

```
Your Data (NumPy / Pandas)
        ↓
Seaborn (optional, high-level API)
        ↓
Matplotlib (low-level rendering engine)
        ↓
Figure → Axes → Artists → Pixels
```

Seaborn decides what to plot and how to summarize the data;
Matplotlib decides how it is drawn and renders it to the screen.

Matplotlib

plt.plot()

- Get current Figure
- Get current Axes
- Create a Line2D artist
- Attach it to Axes
- Axes updates limits
- Renderer draws pixels

```
Figure
 ├── Axes (subplot)
 │     ├── Line2D
 │     ├── PathCollection (scatter)
 │     ├── Patch (bars, boxes)
 │     ├── Text (labels, titles)
 │     └── Legend
```

- **Figure**: The entire canvas (window / image)
- **Axes**: A single plot area (x-axis + y-axis)
- **Artists**: Everything that gets drawn

Seaborn

Instead of:

```
plt.scatter(df["age"], df["salary"])
```

You write:

```
sns.scatterplot(data=df, x="age", y="salary", hue="gender")
```

Seaborn:

- Accepts DataFrames
- Handles grouping, aggregation, statistics
- Maps variables to visual channels automatically

Seaborn – does statistical transformations

Seaborn does data processing before plotting.

Example:

```
sns.barplot(data=df, x="department", y="salary")
```

Behind the scenes:

- Group by department
- Compute mean salary
- Compute confidence intervals
- Pass computed values to Matplotlib

Matplotlib itself does none of this.

Seaborn – does nice styling

Seaborn sets:

- Color palettes
- Grid styles
- Font sizes
- Tick directions
- Background aesthetics

Under the hood:

sns.set_theme()→ modifies Matplotlib rcParams

Seaborn – end to end processing

```
sns.boxplot(data=df, x="department", y="salary")
```

Step 1: Input interpretation
- Detect categorical vs numeric
- Validate column names
- Infer plot semantics

Step 2: Statistical computation
- Group data by department
- Compute quartiles, whiskers, outliers

Step 3: Create Matplotlib objects
- Create a Figure (if not exists)
- Create an Axes
- Create Patch artists for boxes
- Create Line2D for whiskers

Step 4: Styling
- Apply color palette
- Set linewidths
- Configure grid & ticks

Step 5: Render
- Matplotlib renderer converts artists → pixels
- Output to screen or file

Seaborn for plots and stats, matplotlib for fine-grained control

You still use Matplotlib for:

- Custom annotations
- Secondary y-axes
- Subplot grids beyond defaults
- Exact control of ticks, spines, limits
- Exporting figures precisely (DPI, formats)

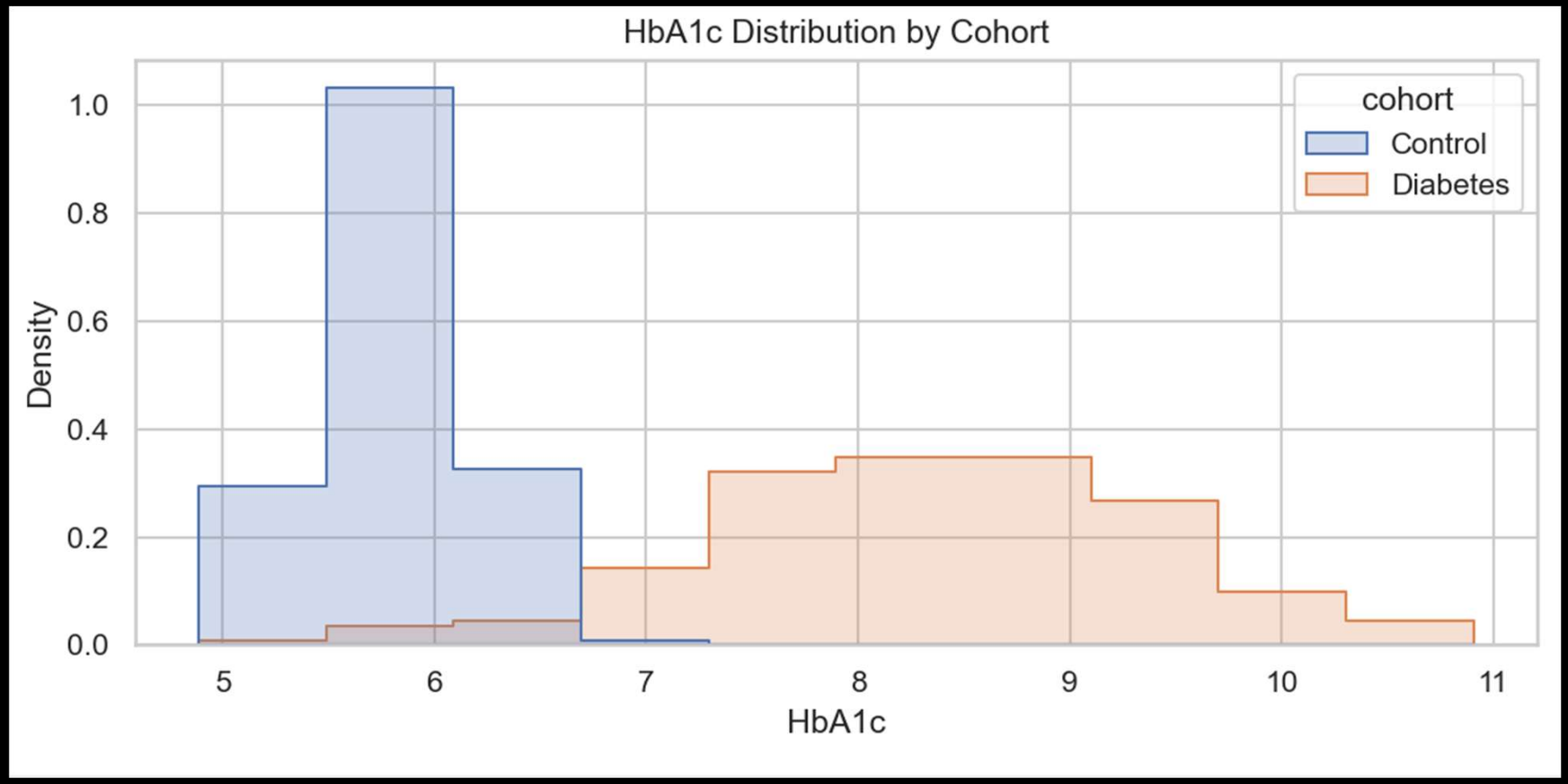
Axis titles

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="whitegrid")

ax = sns.lineplot(data=df, x="date", y="revenue")

ax.set_title("Revenue Trend")
ax.set_ylabel("USD (Millions)")
plt.tight_layout()
plt.show()
```

Draw a vertical line to indicate mean

```
ax = sns.histplot(df["price"], bins=50)
ax.axvline(df["price"].mean(), color="red")
```

Interpret this chart



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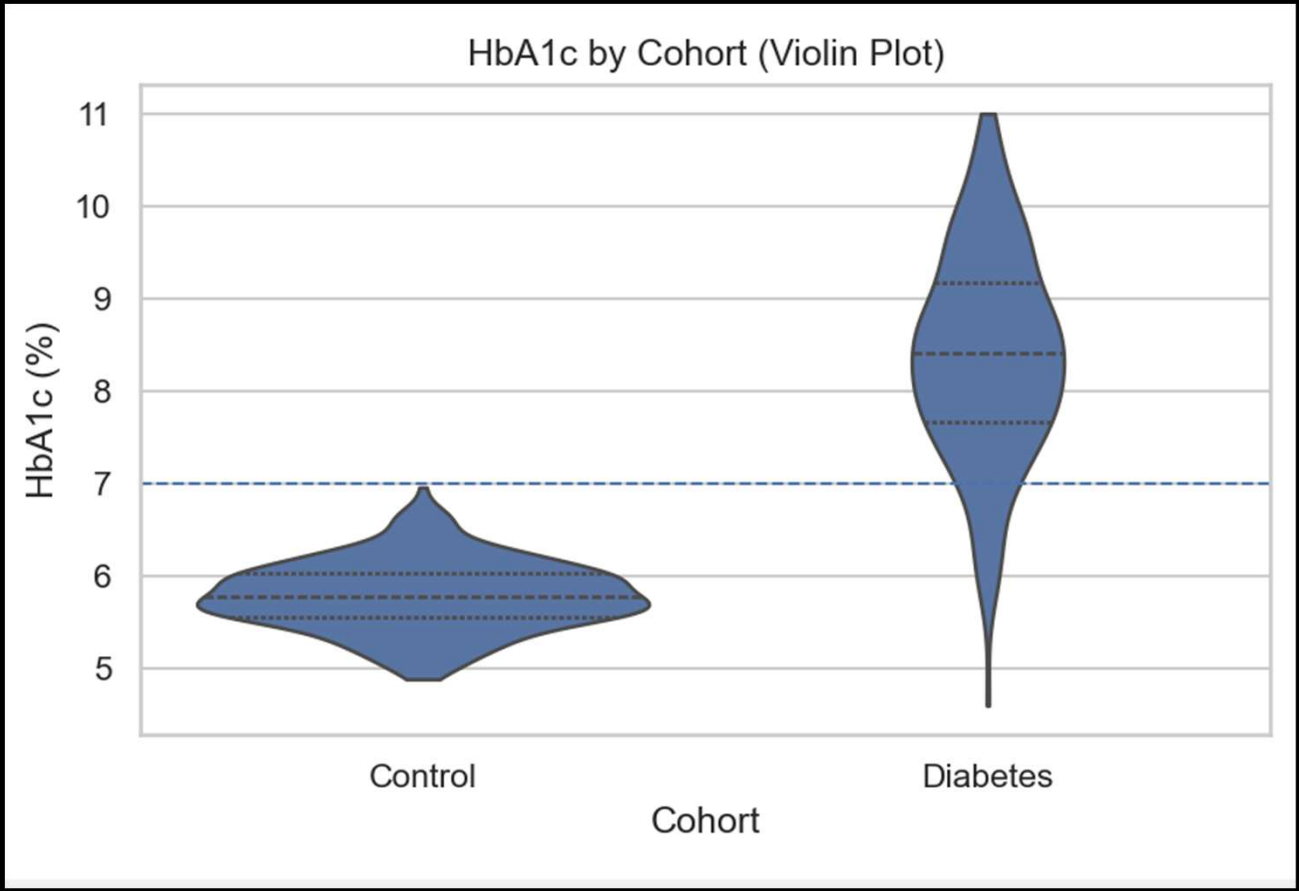

Enterprise Machine Learning with Microsoft Azure

Data ingestion

- **Azure Data Factory**
  Low code/no code tool:
  Visual pipeline designer, control flows, event triggers, scheduling
- Data ingestion, ETL at scale
- Integrates with Databricks, Synapse
- For async http calls, Python asyncio, aiohttp are suitable

Data Eng., model training

- **Azure Databricks**
  data cleaning, feature engineering at scale with Apache Spark
- Distributed Model training with Apache Spark. Integrates with Hadoop
- Notebooks
- Metadata, Delta Tables

Data Storage

- Parquet files in Azure Data Lake or AWS S3
- Azure Synapse
  If you require complex sql queries, reporting or ad-hoc analysis on structured data